

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0510 Slot A

COURSE OUTCOME COVERED

CO5: Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos. (BTL 6)

Assessment Tool Weightage: 10%

QUESTIONS:4 Total Marks 20 Duration :60 Minutes Annexure A
Mock Interview

Code of Conduct:

I M.Venkata Veeranajneya Reddy (192472140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. IL= understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Venkata Veeranajneya Reddy

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Image & Video Processing

1. A video stream drops frames during processing. Analyze the problem and redesign your approach to ensure smooth playback, explaining trade-offs and optimization techniques.

Ans:

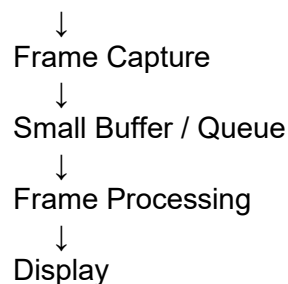
Problem Analysis

Frames are dropping because the **processing time per frame is greater than the time available before the next frame arrives**. If the system takes too long to process each frame, frames accumulate in the buffer and playback becomes delayed or frames are skipped.

Proposed Approach

Use a pipeline that separates **frame capture** from **frame processing**.

Video Stream



The capture thread continuously obtains frames, while the processing stage handles them independently.

Optimization Techniques

1. **Resize frames** before processing to reduce computational cost.
2. **Skip frames** when processing cannot keep up.
3. Use a **small queue** so old frames do not create large delays.
4. Process only the required **ROI** instead of the complete frame.
5. Use lightweight OpenCV algorithms where possible.
6. Avoid unnecessary conversions and repeated operations.
7. If available, use **OpenCV DNN acceleration/GPU**.

Pseudocode

ALGORITHM SmoothVideoProcessing

START

```

INITIALIZE video capture
INITIALIZE small frame queue
  
```

```

WHILE video is available DO

    CAPTURE latest frame

    IF queue is full THEN
        DISCARD oldest frame
    END IF

    ADD latest frame to queue

    IF processing unit is free THEN

        GET latest frame from queue

        RESIZE frame

        PREPROCESS frame

        PERFORM required processing

        DISPLAY processed frame

    END IF

END WHILE

RELEASE video
CLOSE windows

```

END

Trade-offs

- **Frame skipping:** Improves smoothness but reduces temporal accuracy.
- **Lower resolution:** Improves speed but reduces image detail.
- **ROI processing:** Improves performance but ignores information outside the ROI.
- **GPU acceleration:** Improves speed but requires suitable hardware.

Final Decision

For smooth playback, it is better to **process fewer frames at lower resolution than to process every frame slowly**, especially when real-time response is more important than processing every individual frame.

2. You need to process only every 5th frame of a video. Design an efficient solution and justify your approach, considering performance and accuracy.

Ans:

Problem Analysis

Processing every frame wastes computational resources when only periodic analysis is required. The solution is to read all frames sequentially but perform expensive processing only when the frame number is divisible by 5.

Proposed Approach

Frame 0 → Process

Frame 1 → Skip

Frame 2 → Skip

Frame 3 → Skip

Frame 4 → Skip

Frame 5 → Process

Frame 6 → Skip

...

Pseudocode

ALGORITHM ProcessEveryFifthFrame

START

 INITIALIZE video capture

 SET frame_count = 0

 WHILE video is available DO

 READ frame

 IF frame is empty THEN

 BREAK

 END IF

 IF frame_count MOD 5 = 0 THEN

 RESIZE frame if required

 PREPROCESS frame

 PERFORM required processing

 DISPLAY or SAVE result

 END IF

 INCREMENT frame_count

 END WHILE

 RELEASE video

 CLOSE windows

END

Why this is efficient

If a video contains **1000 frames**, only approximately:

$1000 / 5 = 200$ frames

need expensive processing.

Therefore, the computational workload is significantly reduced.

Trade-off

Advantage:

- Lower CPU usage
- Faster processing
- Lower memory/energy consumption
- Suitable when consecutive frames contain similar information

Disadvantage:

- Events occurring between processed frames may be missed.
- Temporal accuracy decreases.

Final Decision

Processing every 5th frame is appropriate when the application does not require analysis of every frame. The modulo operation provides a simple and efficient frame-selection mechanism.

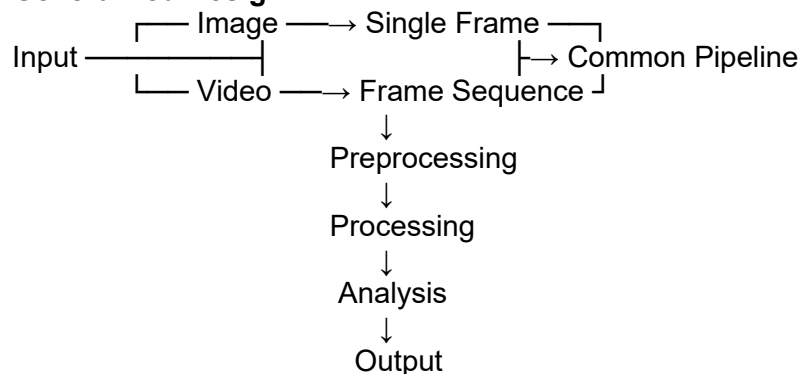
3.A system must process both images and videos using the same pipeline. Propose a generalized design and explain how it handles both cases effectively.

ANS:

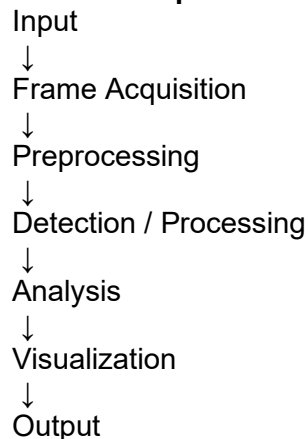
Problem Analysis

Images contain only **one frame**, while videos contain a **sequence of frames**. Therefore, the processing stages should be common, while only the input acquisition stage changes.

Generalized Design



Common Pipeline



Pseudocode

ALGORITHM GeneralImageVideoPipeline

START

GET input

IF input is an IMAGE THEN

 READ image

 ADD image as single frame
 to processing pipeline

ELSE IF input is a VIDEO THEN

 INITIALIZE video capture

 WHILE video has frames DO

 READ next frame

 SEND frame to common pipeline

 END WHILE

END IF

END

FUNCTION CommonPipeline(frame)

 RESIZE frame if required

 CONVERT color space if required

 REMOVE noise

 PERFORM required computer vision operation

 ANALYZE result

 DRAW required information

 RETURN processed frame

END FUNCTION

Why this design is effective

The **processing function is shared** by both image and video inputs.

For an image:

Image → CommonPipeline → Output

For a video:

Video → Frame 1 → CommonPipeline
→ Frame 2 → CommonPipeline
→ Frame 3 → CommonPipeline
→ ...

Advantages

- Avoids duplicate code.
- Easier to maintain.
- Same preprocessing and detection logic can be reused.
- New image/video sources can be added easily.
- Video processing becomes repeated execution of the same frame-processing function.

Final Decision

Separate **input handling** from **frame processing**. This provides a generalized pipeline where an image is treated as a single frame and a video is treated as a sequence of frames.

4. You need to extract frames from a video and save only those with significant changes.
Analyze the problem and design a method, explaining your decision criteria and approach.

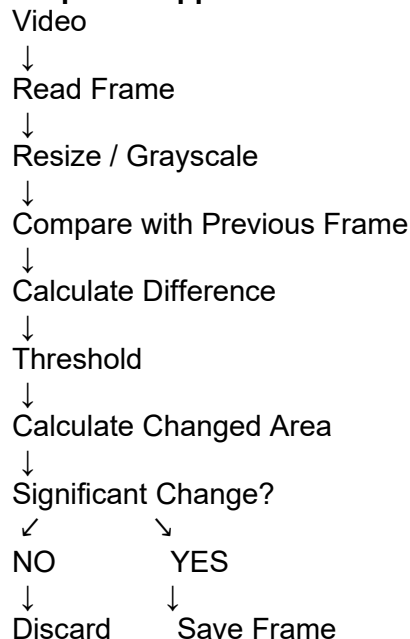
Ans:

Problem Analysis

Saving every video frame requires large storage and produces many redundant images because consecutive frames are often very similar.

Therefore, compare the current frame with the previous/reference frame and save the current frame only when the change is significant.

Proposed Approach



Decision Criteria

Use **frame difference**.

1. Convert current and previous frames to grayscale.
2. Calculate absolute difference:

$\text{Difference} = |\text{Current Frame} - \text{Previous Frame}|$

3. Apply a threshold to identify changed pixels.
4. Calculate the percentage/area of changed pixels.

5. Save the frame if the changed area exceeds a predefined threshold.

For example:

Changed Area < Threshold → Ignore

Changed Area ≥ Threshold → Save

Pseudocode

ALGORITHM SignificantFrameExtraction

START

INITIALIZE video capture

READ first frame

IF first frame is empty THEN

STOP

END IF

CONVERT first frame to grayscale

APPLY Gaussian Blur

SET previous_frame = first frame

WHILE video has frames DO

READ current frame

IF current frame is empty THEN

BREAK

END IF

CONVERT current frame to grayscale

APPLY Gaussian Blur

CALCULATE absolute difference between
current_frame and previous_frame

APPLY threshold to difference image

APPLY morphological operations
to remove small noise

CALCULATE changed pixel area

IF changed area > predefined threshold THEN

SAVE current frame

UPDATE previous_frame
= current frame

END IF

END WHILE

RELEASE video

END

Why use these operations?

- **Grayscale:** Reduces computation because color information is unnecessary for basic change detection.
- **Gaussian blur:** Reduces noise that could appear as false changes.
- **Absolute difference:** Detects changes between consecutive frames.
- **Thresholding:** Separates significant changes from small pixel variations.
- **Morphological filtering:** Removes small noise regions.
- **Area threshold:** Ensures only meaningful changes cause a frame to be saved.

Trade-off

A **low threshold** saves more frames and reduces the chance of missing events, but increases storage.

A **high threshold** saves fewer frames and saves storage, but small important changes may be missed.

Therefore, the threshold should be selected according to the application's requirements.

Final Pipeline

Video Input

